Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc2`; Mathlib at commit `1b1ccf4b`; physlib (`master`) and TauCeti (commit `e7e7f52`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Projective measurements are equilibration processes

Files.

- File: `briefs/ProjectiveMeasurement/ProjectiveMeasurement.lean`
- Library: `ProjectiveMeasurement`
- Imports:
  `ForPhyslib.QuantumMechanics.Dephasing`,
  `ForPhyslib.QuantumMechanics.RelativeEntropy`,
  and the brief `UnitaryEntropy` (for `vonNeumann`)
- New files in `ForPhyslib/QuantumMechanics/`: `Pinching` (spectral projectors as the functional calculus of indicators, the pinching, the commutant, trace duality, density operators and supports, the Hilbert–Schmidt Pythagoras identity), `Dephasing` (the Lindblad generator, the explicit solution, uniqueness, the limit), `RelativeEntropy` (relative entropy, Klein's inequality with its equality case, the relative-entropy Pythagoras identity). They use from TauCeti:
  - the Hilbert–Schmidt norm, `TauCeti.Analysis.InnerProductSpace.HilbertSchmidt.Basic`;

  - finiteness of the spectrum, `TauCeti.Analysis.Normed.Operator.FiniteDimensional`;
  - the eigenvector action of the functional calculus, `TauCeti.Analysis.InnerProductSpace.FunctionalCalculusComparison`;
  - the supporting-line bound for $x \log x$, `TauCeti.Analysis.SpecialFunctions.Log.MulLog`.

- New `ForMathlib` material: `ForMathlib/Analysis/Calculus/MeanValue.lean` (a function continuous from the right at a with derivative zero on (a, ∞) is constant on $[a, \infty)$).

Setting. E is a finite-dimensional complex Hilbert space with density operators as in the brief "Schrödinger equation is equivalent to information conservation". A process is a map Φ sending an observable X and an incoming ensemble ρ to $\Phi(X, \rho)$; the conditions constrain it for self-adjoint X and density operators ρ. For self-adjoint X, `spectralProj` X x is `cfc` of the indicator of $\{x\}$, the projector P_x onto the x-eigenspace, and `pinching` X $\rho = \sum_x P_x \rho P_x$, the sum running over the real spectrum of X. `dephasingGenerator` X $\rho = X\rho X - \frac{1}{2}(X^2\rho + \rho X^2)$. The Hilbert–Schmidt norm is TauCeti's `hilbertSchmidtNorm`. The relative entropy `relEntropy` ρ $\sigma \in [0, \infty]$ is $\operatorname{tr}(\rho \log \rho) - \operatorname{tr}(\rho \log \sigma)$ if $\ker \sigma \subseteq \ker \rho$ and ∞ otherwise, with $\log =$ `cfc Real.log`, which is the logarithm on the support since $\log 0 = 0$. `IsUniqueMinOn` f s a says that $a \in s$ and $f(a) < f(b)$ for every other $b \in s$; `commutingDensities` X is the set of density operators commuting with X. The conditions are

- `EqMeas`: $\Phi(X, \rho) = P_X(\rho)$;
- `EqLind`: every $r : \mathbb{R} \to \operatorname{End}(E)$ with $r(0) = \rho$, $r'(t) = L(r(t))$ for $t > 0$ and r continuous from the right at 0 tends to $\Phi(X, \rho)$;
- `EqHS`: $\Phi(X, \rho)$ is the unique minimizer of $\|\rho - \sigma\|_{HS}$ over `commutingDensities` X;
- `EqKL`: $\Phi(X, \rho)$ is the unique minimizer of $D(\rho \| \sigma)$ over the same set.

Results.

Brief	Lean
Eqs. (3)–(4), blockwise evolution	`spectralProj_mul_dephasingGenerator_mul`, `dephasingGenerator_block`
Eq. (5), the solution	`dephasing`, `hasDerivAt_dephasing`, `dephasing_zero`; uniqueness `eq_dephasing`
Eqs. (6)–(10), the limit	`tendsto_dephasing`, `tendsto_pinching_of_hasDerivAt`
EQ-LIND ⇔ EQ-MEAS	`eqMeas_iff_eqLind`
"$[\sigma, X] = 0$ iff $P_X(\sigma) = \sigma$"	`commute_iff_pinching_eq`
P_X is positive and trace preserving	`isDensity_pinching`
Eq. (11)	`sq_hilbertSchmidtNorm_sub`, `hilbertSchmidtNorm_sub_pinching_lt`
EQ-HS ⇔ EQ-MEAS	`eqMeas_iff_eqHS`
"$\operatorname{tr}(\rho \log \sigma) = \operatorname{tr}(P_X(\rho) \log \sigma)$"	`trace_pinching_mul_of_commute`
Eq. (12)	`relEntropy_eq_add`
Eq. (13)	`relEntropy_pinching`, `relEntropy_pinching_eq`
"non-negative and vanishes iff $\sigma = P_X(\rho)$" (Klein)	`re_trace_mul_log_sub_log_nonneg`, `re_trace_mul_log_sub_log_eq_zero_iff`, `relEntropy_eq_zero_iff`
Supports in eq. (12)	`ker_pinching_le`, `ker_le_ker_pinching`
EQ-KL ⇔ EQ-MEAS	`eqMeas_iff_eqKL`, via `relEntropy_pinching_lt`
Theorem 1, complete	`projective_measurement_tfae`

Differences from the brief.

- Finite dimension only; the brief has a separable Hilbert space. Then every self-adjoint X has pure point spectrum, every density operator has finite entropy, so [B]FINENT holds and the two halves of Theorem 1 merge into one equivalence of four conditions. The convergence in EQ-LIND is in operator norm, which in finite dimension is trace-norm convergence; the non-commutative Scheffé lemma and dominated convergence are not needed.

- The Lindblad equation is assumed only for $t > 0$, with continuity from the right at $t = 0$. Under these assumptions the solution is unique (`eq_dephasing`), which the brief takes for granted when it solves eq. (4). EQ-LIND is stated for every solution; the explicit one shows that there is one.

- The Hilbert–Schmidt Pythagoras identity and the strict inequality hold for every operator σ commuting with X, not only for density operators; the constraint $\sigma \in D(H)$ in EQ-HS is not needed. For EQ-KL it is, through Klein's inequality.

- Eq. (13) is stated with the entropy `vonNeumann` of the brief "Schrödinger equation is equivalent to information conservation" and the identity $S(\rho) = -\operatorname{tr}(\rho \log \rho)$ (`vonNeumann_eq_neg_re_trace`). As a value in $[0, \infty]$ it is $\max(S(P_X(\rho)) - S(\rho), 0)$, which equals $S(P_X(\rho)) - S(\rho)$ by Klein's inequality.

- The relative entropy takes values in $[0, \infty]$, as physlib's `qRelativeEnt`. physlib's pinching and `pinching_pythagoras` are stated for matrices and pinch by the eigenprojectors of a state; this formalization works with operators on E, as the other briefs do, and pinches by an observable.

Problems found in the brief. No errors in the statements. Gaps and implicit points:

- Klein's inequality, "the last term is non-negative and vanishes if and only if $\sigma = P_X(\rho)$", is used without a reference. It is proved here in eigenbases: with $c_{ij} = |\langle u_i|v_j\rangle|^2$, $D(\rho\|\sigma) = \sum_{ij} c_{ij}(r_i \log r_i - r_i \log s_j - r_i + s_j) \geq \sum_{ij} c_{ij}(\sqrt{r_i} - \sqrt{s_j})^2$.
- "For non-faithful σ, the identity is understood on its support" hides two support facts that eq. (12) needs: $\ker P_X(\rho) \subseteq \ker \rho$ for positive ρ, and $\ker \sigma \subseteq \ker \rho \Rightarrow \ker \sigma \subseteq \ker P_X(\rho)$ for σ commuting with X. The first also gives eq. (12) when $\ker \sigma \not\subseteq \ker \rho$: then both sides are infinite.
- "Both $\rho(t)$ and $P_X(\rho(0))$ are density operators": that $\rho(t)$ stays a density operator is asserted, not shown. It holds because the matrix $\left(e^{-t(x-y)^2/2}\right)_{x,y}$ is positive semidefinite, so by the Schur product theorem the blockwise multiplication preserves positivity. In infinite dimension the Scheffé step needs it; in finite dimension it is not needed and is not formalized.
- The sentence "This base condition states the ensembles have finite entropy" follows QSTO:H, which does not say this; it describes [B]FINENT, stated next.
- The remark on observables with continuous spectrum is not formalized; in finite dimension there is none.

1.1 Hilbert spaces of any dimension

This subsection covers Theorem 1 for bounded observables with pure point spectrum: EQ-MEAS, EQ-LIND and EQ-HS are equivalent, and under [B]FINENT EQ-KL is equivalent to EQ-MEAS. It also covers the claims of the conclusion on repeatability and state update, the remark that the conditions fail on a continuous part of the spectrum, and Theorem 1 for unbounded observables with pure point spectrum.

Files.

- File: `briefs/ProjectiveMeasurement/ProjectiveMeasurement/InfiniteDimensional.lean`
- Library: `ProjectiveMeasurement`, which now builds every module `ProjectiveMeasurement.*`
- Imports:
  `ForPhyslib.QuantumMechanics.DephasingFlow`,
  `ForPhyslib.QuantumMechanics.VonNeumannRelativeEntropy`,
  and the finite-dimensional `ProjectiveMeasurement` (for `IsUniqueMinOn`)
- New file `ForPhyslib/QuantumMechanics/DiscreteObservable.lean`: resolutions of the identity and observables with pure point spectrum, the pinching as a strong sum, blocks and block-diagonal operators, positivity and trace of the pinching, and the Hilbert–Schmidt Pythagoras identity, through TauCeti's block splitting of the Hilbert–Schmidt energy (`TauCeti.Analysis.InnerProductSpace.HilbertSchmidt.Pythagoras`).
- New file `ForPhyslib/QuantumMechanics/DephasingFlow.lean`: the bounded generator and its exponential, the blocks of a solution, the unitary group e^{isX}, the Gaussian average and its trace-norm contraction, equilibration in trace norm.

- New file `ForPhyslib/QuantumMechanics/VonNeumannRelativeEntropy.lean`: eigenbases, the entropy as $\sum_i \eta(\lambda_i)$, the cross term $-\operatorname{tr}(\rho \log \sigma)$, Lindblad's formula, Klein's inequality, and the behaviour under pinching.

- New file `ForMathlib/Analysis/InnerProductSpace/CFCEigenvector.lean`: $g(T)v = g(\mu)v$ for $Tv = \mu v$, T self-adjoint and g continuous on the spectrum, in any dimension (TauCeti's version is finite-dimensional).

- Infinite distance when $S(P_X(\rho)) = \infty$, in a module of its own, so that the comparator lane is unchanged:
  `briefs/ProjectiveMeasurement/ProjectiveMeasurement/InfiniteDistance.lean`
  Imports: `ProjectiveMeasurement.InfiniteDimensional` and
  `ForPhyslib.QuantumMechanics.VonNeumannEntropyMixture`
  (the latter for a density operator of infinite entropy).

- Agreement of the two relative entropies, in a module of its own, so that the comparator lane is unchanged:
  `briefs/ProjectiveMeasurement/ProjectiveMeasurement/RelativeEntropies.lean`
  Imports: `ProjectiveMeasurement.InfiniteDimensional` and
  `ForPhyslib.QuantumMechanics.RelativeEntropyAgreement`

- New file `ForPhyslib/QuantumMechanics/RelativeEntropyAgreement.lean`: in finite dimension, `relativeEntropy` and `relEntropy` agree on density operators, and every entropy is finite.

- New file:
  `ForPhyslib/QuantumMechanics/LindbladSumBasisIndependence.lean`
  Lindblad's formula in terms of the eigenspace projections, so that it does not depend on the eigenbases.

- The bridge from the spectral theorem, in a module of its own, so that the comparator lane is unchanged:
  `briefs/ProjectiveMeasurement/ProjectiveMeasurement/SpectralTheorem.lean`
  Imports: `ProjectiveMeasurement.InfiniteDimensional` and
  `ForPhyslib.QuantumMechanics.PurePointSpectralMeasure`

- New file:
  `ForPhyslib/QuantumMechanics/PurePointSpectrum.lean`
  eigenspace projections `eigenproj`, pure point spectrum, and the existence and uniqueness of eigendecompositions.

- New file:
  `ForPhyslib/QuantumMechanics/PurePointSpectralMeasure.lean`
  the spectral projections of points are the eigenspace projections, for the spectral measure `boundedSelfAdjointSpectralMeasure` of a bounded self-adjoint operator in PhyslibAlpha.

- Repeatability and state update, in a module of its own, so that the comparator lane is unchanged:
  `briefs/ProjectiveMeasurement/ProjectiveMeasurement/Repeatability.lean`
  Imports: `ProjectiveMeasurement.InfiniteDimensional` and
  `ForPhyslib.QuantumMechanics.LuedersRule`

- New file `ForPhyslib/QuantumMechanics/LuedersRule.lean`: outcome probabilities $p_x = \mathrm{tr}(P_x \rho P_x)$, the Lüders state $\rho_x = P_x \rho P_x / p_x$, repeatability, and conditioning of mixtures.

- EQ-LIND by the route of the brief, eqs. (6)–(10), in a module of its own, so that the comparator lane is unchanged:
  `briefs/ProjectiveMeasurement/ProjectiveMeasurement/HilbertSchmidtRoute.lean`
  Imports: `ProjectiveMeasurement.InfiniteDimensional` and `ForPhyslib.QuantumMechanics.DephasingHilbertSchmidt`

- New file:
  `ForPhyslib/QuantumMechanics/NoncommutativeScheffe.lean`
  trace-norm bounds averaged over the spectral expansion, the gentle measurement bound, and the non-commutative Scheffé lemma for weak and for Hilbert–Schmidt convergence.

- New file:
  `ForPhyslib/QuantumMechanics/DephasingHilbertSchmidt.lean`
  solutions of the dephasing equation stay density operators, their Hilbert–Schmidt distance to $P_X(\rho)$, its convergence by dominated convergence (Mathlib's Tannery theorem, `Mathlib.Analysis.Normed.Group.Tannery`), and trace-norm convergence through the Scheffé lemma.

- Observables with continuous spectra, in a module of its own, so that the comparator lane is unchanged:
  `briefs/ProjectiveMeasurement/ProjectiveMeasurement/ContinuousSpectrum.lean`
  Imports: `ProjectiveMeasurement.InfiniteDimensional` and `ForPhyslib.QuantumMechanics.ContinuousSpectrum`

- New file:
  `ForPhyslib/QuantumMechanics/ContinuousSpectrum.lean`
  the point-spectral subspace, compact operators commuting with X, spectral sectors without atoms, the equilibria of the dephasing equation for every bounded self-adjoint X, and limits of the dephasing flow.

- New file:
  `ForMathlib/Analysis/CStarAlgebra/Commutator.lean`
  for a self-adjoint x in a C*-algebra, $[x,[x,a]] = 0$ implies $[x,a] = 0$.

- New file:
  `ForMathlib/Analysis/Calculus/Deriv/Tendsto.lean`
  if f converges at $+\infty$ and $f' \to c$, then $c = 0$.

- Unbounded observables, in a module of its own, so that the comparator lane is unchanged:
  `briefs/ProjectiveMeasurement/ProjectiveMeasurement/Unbounded.lean`
  Imports: `ProjectiveMeasurement.InfiniteDimensional`, `ForPhyslib.QuantumMechanics.UnboundedDephasing` and `PhyslibAlpha.ProbabilisticTheory.HilbertSpace.Unbounded.CayleySpectralData.SpecTheorem`

- New file:
  `ForPhyslib/QuantumMechanics/UnboundedDephasing.lean`
  Gaussian averages over a strongly continuous phase group, with their blocks, the trace-norm

contraction, the semigroup property and equilibration; for a self-adjoint T with atomic PhyslibAlpha spectral measure, the eigenprojections, e^{isT}, T on spectral windows, and the dephasing semigroup as the unique extension from the windows.

Setting. E is a complex Hilbert space of any dimension; separability is not used. Density operators, pure states and the entropy `vonNeumannEntropy` $\in [0, \infty]$ are as in the section "Hilbert spaces of any dimension" of `briefs/SchroedingerEvolution/formalization-log.tex`. An observable is $X \in \mathrm{End}(E)$ with `IsEigenDecomposition` X P x: the P_x are pairwise orthogonal projections with $\sum_x P_x = 1$ strongly, the x are distinct reals, and $XP_x = xP_x$; the index set is arbitrary. The pinching `IsResolutionOfIdentity.pinching` $\rho = \sum_x P_x \rho P_x$ is a strong sum, defined for every bounded ρ. The conditions are those of finite dimension, with these changes:

- `EqLind`: solutions are differentiable in operator norm for $t > 0$ and continuous from the right at 0; they tend to $\Phi(X, \rho)$ in trace norm.
- `EqHS`: minimization over `commutingDensityOperators` X, with TauCeti's `hilbertSchmidtNorm`.
- `EqKL`: only for ρ with $S(\rho)$ and $S(P_X(\rho))$ finite, and minimization over the density operators of finite entropy commuting with X. `EqMeasFinEnt` is EQ-MEAS on the same ρ.
- The relative entropy `relativeEntropy` is Lindblad's formula $\sum_{ij} |\langle e_i | f_j \rangle|^2 (\lambda_i \log \lambda_i - \lambda_i \log \mu_j + \mu_j - \lambda_i)$ in eigenbases of ρ and σ chosen once, each term in $[0, \infty]$ and $+\infty$ for $\mu_j = 0 < \lambda_i$.

Results.

Brief	Lean
Eq. (2), blocks determine ρ	`IsResolutionOfIdentity.ext_block`
Eqs. (3)–(4)	`IsEigenDecomposition.mul_dephasingGenerator_mul`
Eq. (5), for every solution	`IsEigenDecomposition.mul_eq_of_hasDerivAt`
A solution exists	`dephasingFlow`, `hasDerivAt_dephasingFlow`
$\rho(t)$ is a mixture of unitary conjugations of $\rho(0)$	`gaussianDephasing`, `IsEigenDecomposition.mul_gaussianDephasing_mul`, `IsEigenDecomposition.eq_gaussianDephasing`
Eqs. (6)–(10), trace-norm convergence	`IsEigenDecomposition.traceENorm_gaussianDephasing_sub_le`, `IsEigenDecomposition.tendsto_traceENorm_of_hasDerivAt`
"Both $\rho(t)$ and $P_X(\rho(0))$ are density operators"	`IsEigenDecomposition.gaussianDephasing_nonneg`, `IsEigenDecomposition.isDensityOperator_of_hasDerivAt`
Eqs. (6)–(9), the brief's route: Hilbert–Schmidt convergence	`IsEigenDecomposition.hilbertSchmidtEnergy_sub_pinching_of_hasDerivAt`, `IsEigenDecomposition.tendsto_hilbertSchmidtNorm_of_hasDerivAt`

Brief	Lean
Non-commutative Scheffé lemma, between eqs. (9) and (10)	`tendsto_traceENorm_of_tendsto_hilbertSchmidtNorm`, for weak convergence `tendsto_traceENorm_of_tendsto_inner`; gentle measurement `traceENorm_sub_finiteProj_le`
Eq. (10) and EQ-LIND ⇔ EQ-MEAS, the brief's route	`IsEigenDecomposition.tendsto_traceENorm_of_hasDerivAt'`, `eqMeas_iff_eqLind'`
EQ-LIND ⇔ EQ-MEAS	`eqMeas_iff_eqLind`
"$[\sigma, X] = 0$ iff block diagonal iff $P_X(\sigma) = \sigma$"	`IsEigenDecomposition.commute_iff`, `IsResolutionOfIdentity.pinching_eq_self_iff`
P_X is positive and trace preserving	`IsResolutionOfIdentity.pinching_nonneg`, `posTrace_pinching`, `isDensityOperator_pinching`
Eq. (11)	`IsResolutionOfIdentity.hilbertSchmidtEnergy_sub`, `hilbertSchmidtNorm_sub_pinching_lt`
EQ-HS ⇔ EQ-MEAS	`eqMeas_iff_eqHS`
$-\operatorname{tr}(\rho \log \sigma)$	`crossEntropy`, `IsEigenbasis.crossEntropy_eq`
Bridge from the spectral theorem: pure point spectrum gives $X = \sum_x x\, \mu_S(\{x\})$	`IsSelfAdjoint.isEigenDecomposition_eigenproj`, `isEigenDecomposition_spectralMeasureProj`, `isEigenDecomposition_spectral`
$\mu_S(\{x\})$ is the eigenspace projection, and $P_x = \mu_S(\{x\})$ in every eigendecomposition	`spectralMeasureProj_singleton`, `IsEigenDecomposition.eigenproj_eq`, `IsEigenDecomposition.spectralMeasureProj_singleton_eq`
Pure point spectrum iff $\sum_x \mu_S(\{x\}) = 1$ strongly	`hasPurePointSpectrum_iff_hasSum_eigenproj`, `hasPurePointSpectrum_iff_hasSum_spectralMeasureProj`, `IsEigenDecomposition.hasPurePointSpectrum`
P_X depends on X only; EQ-MEAS with $P_x = \mu_S(\{x\})$	`IsEigenDecomposition.pinching_eq_pinching`, `eqMeas_iff_spectral`
$D(\rho\|\sigma)$ does not depend on the eigenbases	`IsEigenbasis.lindbladSum_eq_lindbladSum`, `relativeEntropy_eq_lindbladSum_of_isEigenbasis`, `relativeEntropy_eq_tsum_eigenproj`
$D(\rho\|\sigma) = -\operatorname{tr}(\rho \log \sigma) - S(\rho)$	`crossEntropy_eq_lindbladSum_add`, `relativeEntropy_add_vonNeumannEntropy`, `relativeEntropy_eq_lindbladSum`
"$\operatorname{tr}(\rho \log \sigma) = \operatorname{tr}(P_X(\rho) \log \sigma)$"	`IsResolutionOfIdentity.crossEntropy_pinching`
Eq. (12)	`IsResolutionOfIdentity.relativeEntropy_eq_add`
Eq. (13)	`IsResolutionOfIdentity.relativeEntropy_pinching`, `relativeEntropy_pinching_eq`
Klein, "vanishes iff $\sigma = P_X(\rho)$"	`eq_of_lindbladSum_eq_zero`, `relativeEntropy_eq_zero_iff`
EQ-KL ⇔ EQ-MEAS under [B]FINENT	`eqKL_iff_eqMeasFinEnt`, via `IsResolutionOfIdentity.relativeEntropy_pinching_lt`
Theorem 1	`projective_measurement_tfae` and `eqKL_iff_eqMeasFinEnt`

Brief	Lean
$D(\rho\|\sigma) \geq S(P_X(\rho)) - S(\rho)$ for $[\sigma, X] = 0$	`IsResolutionOfIdentity.vonNeumannEntropy_pinching_le`
"Every ensemble commuting with X is infinitely distant from ρ", for $S(\rho) < \infty$	`relativeEntropy_eq_top_of_vonNeumannEntropy_pinching_eq_top`
The same, for $S(\rho) = \infty$: false	`not_forall_relativeEntropy_eq_top`, `isEigenDecomposition_zero`
In finite dimension, the two relative entropies agree	`relativeEntropy_eq_relEntropy`, `IsDensityOperator.vonNeumannEntropy_ne_top`
In finite dimension, EQ-KL is the same minimization in both parts	`commutingDensities_eq`, `eqKL_iff_relativeEntropy`
Sec. 5, "all outputs are equilibria": $L(\sigma) = 0$ iff $[\sigma, X] = 0$, and such σ is its own output	`dephasingGenerator_eq_zero_iff`, `EqLind.apply_eq_self`
Sec. 5, repeatability: $\Phi(X, \Phi(X, \rho)) = \Phi(X, \rho)$, and the outcome probabilities are unchanged	`EqLind.apply_apply`, `EqMeas.outcomeProb_apply`, `IsResolutionOfIdentity.tsum_outcomeProb_eq_one`
Sec. 5, after the outcome x the state ρ_x is an equilibrium and a second measurement gives x with certainty	`IsResolutionOfIdentity.isDensityOperator_luders`, `commute_luders`, `IsResolutionOfIdentity.outcomeProb_luders`
Sec. 5, "the output is a classical statistical mixture": $\Phi(X, \rho) = \sum_x p_x \rho_x$	`IsResolutionOfIdentity.hasSum_pinching_apply_luders`, `EqMeas.hasSum_apply_luders`
Sec. 5, "the update ... is equivalent to what happens in the classical case"	`IsResolutionOfIdentity.luders_of_hasSum`, `EqMeas.luders_apply_eq_component`, `EqMeas.luders_apply`
Sec. 4, a compact operator commuting with X, in particular a density operator, lives on the closed span H_{pp} of the eigenvectors	`apply_mem_pointSubspace_of_commute`, `apply_mem_pointSubspace_of_isTraceClass`, `apply_mem_pointSubspace_of_mem_commutingDensityOperators`
Sec. 4, an operator "supported on a non-atomic spectral sector and commuting with X ... cannot be a non-zero trace-class operator"	`eq_zero_of_spectralMeasureProj_mul_eq`, `not_mem_commutingDensityOperators_of_spectralMeasureProj_mul_eq`
Sec. 4, Lindblad fixed points: $L(\sigma) = 0$ iff $[\sigma, X] = 0$, for every bounded self-adjoint X, so density-operator fixed points live on H_{pp}	`IsSelfAdjoint.commute_of_commute_commutator`, `dephasingGenerator_eq_zero_iff_commute`, `apply_mem_pointSubspace_of_dephasingGenerator_eq_zero`
Sec. 4, trace-norm limits of solutions of the Lindblad equation are fixed points	`eq_zero_of_tendsto_of_hasDerivAt`, `dephasingGenerator_eq_zero_of_tendsto_traceENorm`, `apply_mem_pointSubspace_of_tendsto_traceENorm`
Sec. 4, "all four conditions fail when X has purely continuous spectrum"	`not_isEigenDecomposition`, `commutingDensityOperators_eq_empty`, `not_isUniqueMinOn_commutingDensityOperators`, `not_exists_isDensityOperator_dephasingGenerator_eq_zero`, `not_tendsto_traceENorm`

Brief	Lean
Sec. 3, unbounded X: the eigenprojections $\mu_S(\{a\})$ of an atomic spectral measure project onto eigenvectors, and commuting with X means commuting with them	`apply_pointProj_apply`, `IsAtomic.isResolutionOfIdentity`, `IsAtomic.commute_iff`
Sec. 3, "if X is unbounded, the differential equation is initially valid only on the domain of its generator": on the spectral window $[-m, m]$, X is bounded and the semigroup solves the equation	`apply_spectralCutoffProj_apply`, `IsAtomic.hasDerivAt_dephasingSemigroup`, `IsAtomic.dephasingFlow_eq`
Sec. 3, "the blockwise solution extends the resulting semigroup to every density operator"	`IsPhaseGroup.mul_gaussianConj_mul`, `IsAtomic.isDephasingExtension`, `IsAtomic.eq_of_isDephasingExtension`, `exists_unique_extension`, `IsAtomic.dephasingSemigroup_add`
Sec. 3, $\rho(t)$ stays a density operator, for unbounded X	`IsPhaseGroup.isDensityOperator_gaussianConj`, `isDensityOperator_dephasingSemigroup`
Theorem 1 for unbounded X: EQ-LIND $\Leftrightarrow$ EQ-MEAS $\Leftrightarrow$ EQ-HS, and EQ-KL $\Leftrightarrow$ EQ-MEAS under [B]FINENT	`IsPhaseGroup.tendsto_traceENorm_gaussianConj`, `Unbounded.eqMeas_iff_eqLind`, `Unbounded.eqMeas_iff_eqHS`, `Unbounded.eqKL_iff_eqMeasFinEnt`, `Unbounded.projective_measurement_tfae`

Differences from the brief.

- Observables are bounded with pure point spectrum, given by their eigenprojections and distinct eigenvalues. The brief allows any self-adjoint X and remarks that the conditions fail on a continuous part of the spectrum, which is formalized separately (see below), as are unbounded observables with pure point spectrum. The bridge from the spectral theorem is formalized: a bounded self-adjoint X has pure point spectrum (its eigenvectors span a dense subspace) iff its spectral measure μ_S is atomic, $\sum_x \mu_S(\{x\}) = 1$ strongly, and then $X = \sum_x x\,\mu_S(\{x\})$. Every eigendecomposition has $P_x = \mu_S(\{x\})$, the projection onto $\ker(X - x)$, so the pinching and the conditions depend on X only. The identification uses $\|(X-a)v\|^2 = \int |r-a|^2\, d\langle v, \mu_S(r)v\rangle$, through PhyslibAlpha's bounded spectral integral.

- EQ-LIND is proved twice. The first proof (`eqMeas_iff_eqLind`) uses neither the Hilbert–Schmidt estimate (6)–(9) nor the non-commutative Scheffé lemma. Every solution has the blocks of eq. (5), so it equals the Gaussian average $\int e^{isX}\rho\, e^{-isX}\, dN(0,t)(s)$, whose blocks carry the characteristic function $e^{-t(x-y)^2/2}$. As an average of unitary conjugations, computed in $\mathcal{S}_1$, it is a trace-norm contraction. With $Q = \sum_{x\in F} P_x$ for a finite set F, $\|\rho(t) - P_X(\rho)\|_1$ is at most $\|\rho - Q\rho Q\|_1 + \|QP_X(\rho)Q - P_X(\rho)\|_1$, both small for large F uniformly in t, plus finitely many off-diagonal blocks of $Q\rho Q$, each decaying like $e^{-t(x-y)^2/2}$.

- The second proof (`eqMeas_iff_eqLind'`) follows the brief. The Hilbert–Schmidt energy of $\rho(t) - P_X(\rho)$ is $\sum_{x\neq y} e^{-t(x-y)^2}\|P_x\rho P_y\|_{HS}^2$, which tends to zero by dominated convergence, the bound being $\|\rho\|_{HS}^2 \leq 1$. The Scheffé lemma is proved in the stronger form of Simon, *Trace Ideals*, Thm. 2.20 for $p = 1$: density operators converging *weakly* to a density operator

converge in trace norm. The proof truncates with $Q = \sum_{k\in F} |b_k\rangle\langle b_k|$, the b_k eigenvectors of the limit ρ carrying weight $> 1-\eta^2$, and uses the gentle measurement bound $\|\sigma - Q\sigma Q\|_1 \leq (1-\operatorname{tr}(Q\sigma))/\eta + \eta$ (cf. Winter 1999, Lemma 9), obtained by averaging the pure-state bound $\||c\rangle\langle c| - |Qc\rangle\langle Qc|\|_1 \leq 2\|c - Qc\|$ over the spectral expansion of σ. That $\rho(t)$ is a density operator follows from the Gaussian representation: the average of positive operators is positive, and its trace is that of its pinching, which is $P_X(\rho)$.

- The relative entropy follows decision D4: Lindblad's formula, in eigenbases chosen once. The cross term $-\operatorname{tr}(\rho\log\sigma)$ is defined without bases, as $\sup_n \operatorname{tr}(g_n(\sigma)\rho\, g_n(\sigma))$ for continuous g_n with $g_n^2 \uparrow -\log$. In any eigenbases, Lindblad's sum plus $S(\rho)$ equals the cross term, so the formula does not depend on the bases once $S(\rho) < \infty$. For every ρ it does not depend on them either: with P_a and Q_b the projections onto the eigenspaces of ρ and σ, grouping the sum by eigenvalues gives $D(\rho\|\sigma) = \sum_{a,b} \operatorname{tr}(Q_b P_a Q_b)\, K(a,b)$ with $K(a,b) = a\log a - a\log b + b - a$ ($+\infty$ for $b = 0 < a$), all terms in $[0,\infty]$ (`relativeEntropy_eq_tsum_eigenproj`). Klein's equality case is read off the sum: a vanishing term forces $\lambda_i = \mu_j$ or $\langle e_i|f_j\rangle = 0$, so ρ and σ agree on every $\langle e_i| \cdot f_j\rangle$. In finite dimension it agrees with the finite-dimensional `relEntropy` on density operators (`relativeEntropy_eq_relEntropy`): the entropy is finite, so Lindblad's formula may be taken in the eigenvector bases, where it is $\operatorname{tr}(\rho(\log\rho - \log\sigma))$ if $\ker\sigma \leq \ker\rho$ and has a term $+\infty$ otherwise. Hence the finite-dimensional EQ-KL minimizes the same function over the same set as the part in any dimension (`eqKL_iff_relativeEntropy`).

- The Pythagoras identity (12) is proved in the form $D(\rho\|\sigma) + S(\rho) = -\operatorname{tr}(P_X(\rho)\log\sigma) = D(P_X(\rho)\|\sigma) + S(P_X(\rho))$. The middle step holds because each $g_n(\sigma)$ commutes with every P_x and the pinching preserves traces. No eigenbasis adapted to the blocks is needed.

- Under [B]FINENT the hypothesis $S(\rho) < \infty$ is redundant: $S(\rho) \leq S(P_X(\rho))$ follows from the identities above. `EqKL` keeps it, as the brief does.

- The Hilbert–Schmidt Pythagoras identity holds in $[0,\infty]$ for every operator, through TauCeti's block splitting of the energy, with no summability assumption. The strict inequality uses that differences of density operators have finite energy.

- The two claims of the conclusion are made precise as follows. The outcome x has the probability $p_x = \operatorname{tr}(P_x\rho P_x)$, and the state after it is the Lüders state $\rho_x = P_x\rho P_x/p_x$; the brief names neither. Repeatability: the equilibria of the dephasing equation are the operators commuting with X, under EQ-LIND each commuting density operator is its own output, so applying the process to its output returns it; ρ_x is again an equilibrium, and gives x with probability one. Classical update: a mixture $\sigma = \sum_x w_x\sigma_x$ of density operators $\sigma_x = P_x\sigma_x P_x$ gives the outcome x with probability w_x, and its Lüders state for x is σ_x. This is conditioning of a classical mixture on the component that occurred. The output $\sum_x p_x\rho_x$ is such a mixture, and its update on x equals the Lüders state of the input ρ.

- The remark on continuous spectra is made precise as follows, for a bounded self-adjoint X with spectral measure μ_S and H_{pp} the closed span of its eigenvectors. A compact operator commuting with X maps into H_{pp}. For a compact self-adjoint ρ, each eigenspace of ρ for an eigenvalue $\lambda \neq 0$ is finite-dimensional and invariant under X, so X diagonalizes on it; these eigenspaces span a dense subspace of the range of ρ. A general compact ρ is $\frac{1}{2}((\rho+\rho^\dagger) + i\cdot i(\rho^\dagger - \rho))$, with both parts self-adjoint and commuting with X. "Supported on a non-atomic spectral sector" is read as $\mu_S(S)\rho = \rho$ with $\mu_S(S\cap\{a\}) = 0$ for every a; then $\mu_S(S)$ vanishes on every eigenvector, since $\mu_S(S)v = \mu_S(S)\mu_S(\{a\})v = \mu_S(S\cap\{a\})v$, hence on H_{pp}, and

$\rho = 0$. The fixed points of the Lindblad equation are the operators commuting with X, for every bounded self-adjoint X: $L(\sigma) = -\frac{1}{2}[X,[X,\sigma]]$, and if $c = [X,\sigma]$ commutes with X, then $e^{itX}\sigma e^{-itX} = \sigma + itc$ has norm $\|\sigma\|$ for all real t, so $c = 0$. A trace-norm limit of a solution is a fixed point, since $\rho'(t)$ tends to L of the limit and $\rho(t+1) - \rho(t) \to 0$. "All four conditions fail" is read, for purely continuous spectrum ($\mu_S(\{a\}) = 0$ for every a), as follows. X has no eigendecomposition, so the pinching is not defined (EQ-MEAS). No density operator commutes with X, so the minimizations of EQ-HS and EQ-KL have no solution. The Lindblad equation has no density-operator fixed point, and no solution converges in trace norm to a density operator (EQ-LIND). The conditions themselves quantify over eigendecompositions only, so they say nothing about such X; the statements above say that no extension to them can hold. In the mixed case, every commuting density operator, every density-operator fixed point and every density-operator limit of the flow lives on H_{pp}.

- Unbounded observables are made precise as follows. An observable is a self-adjoint operator T defined on a subspace of E (Mathlib's `LinearPMap`), whose PhyslibAlpha spectral measure μ_S (`cayleyRealSpectralMeasure`) is atomic, $\sum_a \mu_S(\{a\}) = 1$ strongly. Then $\mu_S(\{a\})$ maps into the domain of T and onto eigenvectors of eigenvalue a. A density operator commutes with T when it commutes with every $\mu_S(S)$, equivalently with every $\mu_S(\{a\})$. "Valid only on the domain of its generator" is read on spectral windows: with $Q_m = \mu_S([-m,m])$, TQ_m is the bounded $X_m = \int 1_{[-m,m]}(r)\, r\, d\mu_S(r)$, by PhyslibAlpha's maximal spectral integral, and for $\rho = Q_m\rho Q_m$ the equation is $\rho' = X_m\rho X_m - \frac{1}{2}\{X_m^2, \rho\}$. "Extends the resulting semigroup" is read as a family S_t that solves these equations on every window, with $S_0\rho = \rho$, and is continuous in trace norm on positive trace-class operators. The Gaussian average $\int e^{isT}\rho\, e^{-isT}\, dN(0,t)(s)$, with PhyslibAlpha's unitary group e^{isT}, is such an extension: its blocks are those of eq. (5), and on a window it is the bounded flow of X_m. Since $s \mapsto e^{isT}$ is only strongly continuous, the integral is taken in the trace class, where conjugation by strongly continuous contractions is continuous. Every extension agrees with it on density operators, because $Q_m\rho Q_m \to \rho$ in trace norm. EQ-LIND for unbounded T asks every extension to converge to $\Phi(T,\rho)$ in trace norm; the other conditions are those above with $\mu_S(\{a\})$ as eigenprojections. The maximal domain of the generator is not described.

Problems found in the brief. No errors in the statements. Gaps beyond those of the finite-dimensional part:

- "Both $\rho(t)$ and $P_X(\rho(0))$ are density operators" is asserted. In infinite dimension it is needed for the Scheffé step. The Gaussian representation above proves it, for bounded X (`IsEigenDecomposition.isDensityOperator_of_hasDerivAt`) and for unbounded X (`isDensityOperator_dephasingSemigroup`).

- For unbounded X, the brief does not say on which domain the equation holds, nor in which sense the extension of the semigroup is determined. With the reading above it is the unique extension continuous in trace norm (`IsAtomic.eq_of_isDephasingExtension`).

- The Scheffé step itself ("Hilbert–Schmidt convergence to a density operator implies trace-norm convergence") is cited without reference or proof. It is a known theorem (Simon, *Trace Ideals*, Thm. 2.20; Dell'Antonio 1967), proved here in its weak-convergence form (`tendsto_traceENorm_of_tendsto_inner`).

- The conclusion's claims on repeatability and state update use the outcome probabilities and the post-measurement state without defining them; see the differences above.

- The remark on continuous spectra is asserted without proof. That the fixed points of the Lindblad equation commute with X is not shown for X with a continuous part, where there are no blocks; it holds by $[X,[X,\sigma]] = 0 \Rightarrow [X,\sigma] = 0$ (`IsSelfAdjoint.commute_of_commute_commutator`).

- The brief solves the block equations (4) but does not say that the solution is unique; the argument behind `IsEigenDecomposition.mul_eq_of_hasDerivAt` gives uniqueness for every solution continuous from the right at 0.

- Eq. (13)'s "finite if and only if $S(P_X(\rho))$ is finite" uses $S(\rho) < \infty$, which the admissible domain provides; outside it the equivalence fails, since a block-diagonal ρ with $S(\rho) = \infty$ has $D(\rho\|P_X(\rho)) = D(\rho\|\rho) = 0$. The remark "Relative entropy in infinite dimensions" says that in the case $S(P_X(\rho)) = +\infty$ "every ensemble commuting with X is infinitely distant from ρ". Read literally, the same ρ is a counterexample, with $\sigma = \rho$: in every infinite-dimensional space, for $X = 0$ with the single eigenprojection 1 and a density operator of infinite entropy (`not_forall_relativeEntropy_eq_top`). The claim holds when $S(\rho) < \infty$, by $D(\rho\|\sigma) + S(\rho) = -\operatorname{tr}(P_X(\rho)\log\sigma) \geq S(P_X(\rho))$ (`relativeEntropy_eq_top_of_vonNeumannEntropy_pinching_eq_top`).